1.2 - Calculate V(G) :

Method 1: V(G) = E − N + 2P

Number of Nodes (N) = 13

Number of Edges (E) = 16

Number of Connected Components (P) = 1

V(G) = E − N + 2P

V(G) = 16 − 13 + 2(1)

V(G) = 5

Method 2: Number of Decision Nodes + 1

Decision Nodes:

D1 – patient_id is None

D2 – appointment_id invalid

D3 – amount is None or amount

≤ 0 D4 – status invalid

Total Decision Nodes = 4

V(G) = 4 + 1 = 5

Method 3: Number of Enclosed Regions

The Control Flow Graph contains 5 regions (including the outside region).

V(G) = 5

Verification

Method 1 = 5

Method 2 = 5

Method 3 = 5

All three methods produce the same cyclomatic complexity.

Explanation:

The cyclomatic complexity of create_invoice() is 5.

Therefore, 5 independent basis paths are required for complete basis path testing.

2.1 - List All Basis Paths :

Path #	Node Sequence	Scenario	Feasible?
P1	N1→N2→N3→N13	patient_id is None, so create_invoice() raises a ValueError for a missing patient_id.	Yes
P2	N1→N2→N4→N5→N13	patient_id is valid, but appointment_id is invalid (non-positive or non-integer).	Yes
P3	N1→N2→N4→N6→N7→N13	patient_id and appointment_id are valid, but amount is None or less than or equal to zero.	Yes
P4	N1→N2→N4→N6→N8→N9→N13	All previous inputs are valid, but status is not in VALID_INVOICE_STATUSES.	Yes
P5	N1→N2→N4→N6→N8→N10→N11→N12→N13	All inputs are valid. The invoice is inserted into the database, committed, and the new invoice_id is returned.	Yes

3.1 - DU Pairs for Variable 'record'

Pair #	Def Node	Use Node	Purpose of Use	Def-Clear Path?
DU1	N2	N3	Check whether record is None (record exists).	Yes
DU2	N2	N5	Use record.provider_id to verify the provider is authorized	Yes
DU3	N2	N11	Use record information when creating the prescription.	Yes

Explanation:

The variable record is defined at Node 2 when the medical record is retrieved. It is first used at Node 3 to verify that the record exists, then at Node 5 to compare the provider ID, and finally on the successful execution path at Node 11 when creating the prescription. Since the variable is not redefined between these uses, each path is def-clear.

3.2 – DU Pairs for Variable 'rx':

Pair #	Def Node	Use Node	Purpose of Use	Def-Clear Path?
DU1	N11	N12	Pass rx to db.save(rx) to save the prescription.	Yes
DU2	N11	N13	Return rx.id to the caller after successful creation.	Yes

Explanation:

The variable rx is defined at Node 11 when the Prescription object is created. It is first used at Node 12 when the object is saved using db.save(rx). Finally, it is used at Node 13 when rx.id is returned. Because the variable is never redefined between these uses, both DU pairs follow a def-clear path.

Pytest Verification:

Test: test_du_rx_id_returned_correctly()

Purpose: Verify that the DU pair Def:N11 → Use:N13 is exercised by confirming that rx.id is returned correctly.

Verification Result: pytest tests/test_hw4_du.py -v → 4 passed.